

Technical Report: Final Project DS 5110:
Creating a Data Visualization Dashboard for
Monitoring Key Metrics and Performance Indicators
in a Manufacturing Plant.

Team Members: Qingyuan Hu, Yunmu Shu
Khoury College of Computer Sciences
Data Science Program
hu.qingy@northeastern.edu
shu.yun@northeastern.edu

August 3, 2025

Contents

1	Introduction	2
2	Literature Review	3
3	Methodology	5
3.1	Data Collection	5
3.2	Data Preprocessing	6
3.3	Analysis Techniques	7
4	Results	8
4.1	Dashboards Overview	8
4.2	User Interactions	9
4.3	Accident Forecasting with Machine Learning	10
5	Discussion	11
6	Conclusion	11
7	References	12

1 Introduction

In today’s data-rich environments, traditional reporting methods like static tables or flat charts are no longer sufficient for comprehending complex systems. Interactive data visualization dashboards offer a powerful solution by enabling users to explore large datasets dynamically and intuitively (Figure 1). These dashboards facilitate real-time monitoring, pattern discovery, and anomaly detection. They help decision-makers, engineers, and analysts to gain actionable insights more efficiently, especially when dealing with time-series or multidimensional data, like those commonly found in energy systems or industrial operations.

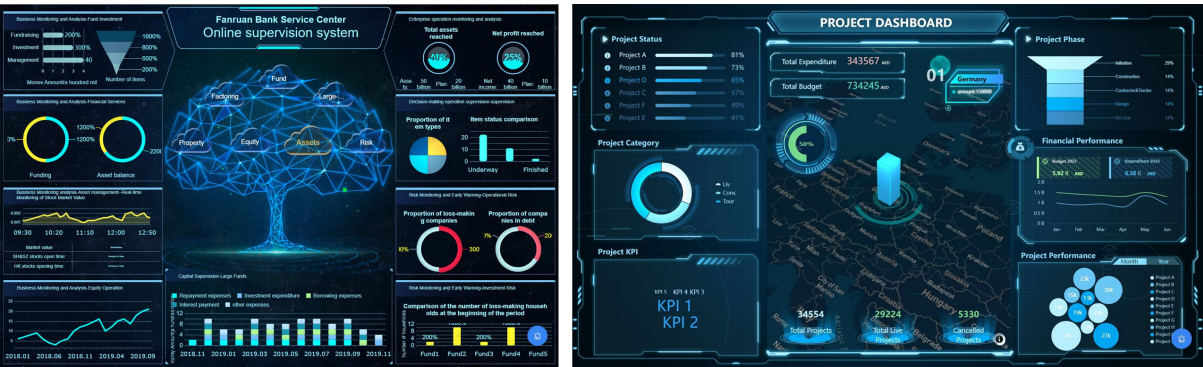


Figure 1: Examples of interactive data visualization dashboards (finereport 2025)

Smart grids (Figure 2) are at the forefront of modern energy systems, integrating renewable energy sources, optimizing load distribution, and ensuring efficient, reliable power delivery. Monitoring smart grid performance is vital due to the fluctuating nature of renewable sources and increasing demand patterns. On the other hand, industrial safety (Figure 3) remains a critical concern, where early identification of potential hazards can prevent costly accidents and ensure operational continuity. Both domains generate continuous, sensor-based data streams—making them ideal use cases for visualization dashboards that enhance situational awareness and proactive decision-making.

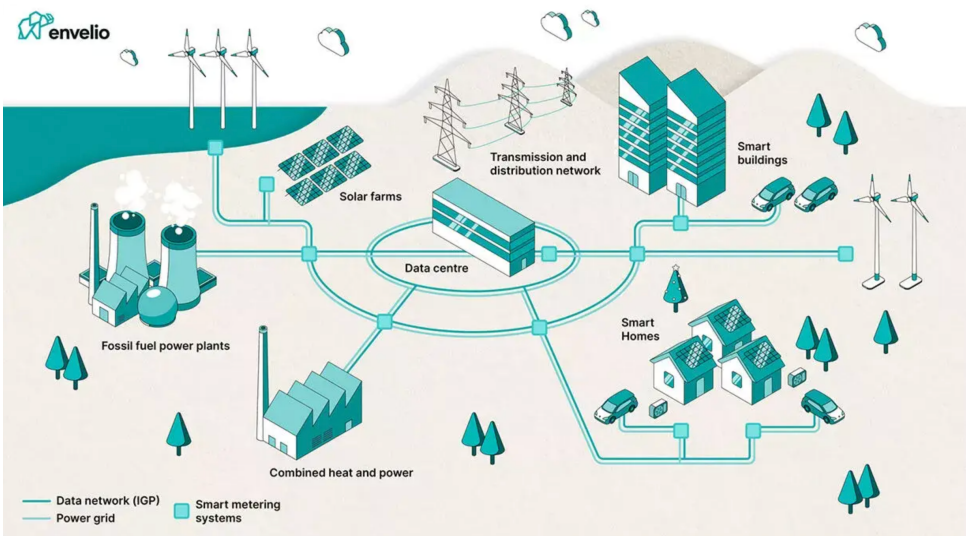


Figure 2: The smart grid connecting key players in the energy system (envelio 2025)



Figure 3: Industrial hygiene and occupational health (btsconsultant 2025)

The primary objective of this project is to design and implement interactive data visualization dashboards for two critical domains: smart grid monitoring and industrial safety analysis. By leveraging the Plotly Dash framework, we aim to provide users with intuitive and dynamic visual interfaces that support real-time monitoring, historical data exploration, and actionable insights. These dashboards transform complex datasets into interpretable formats using time-series plots, pie charts, bar charts, word clouds, and scrollable tables.

The scope of the project includes two datasets: one representing smart grid operations with variables like voltage, current, power consumption, renewable energy input, faults, and environmental conditions; the other focused on workplace incidents with categorical and textual data about accident causes. The dashboards enable users to filter data, detect anomalies, identify patterns, and explore specific scenarios interactively, ultimately supporting data-driven decision-making in both energy management and occupational safety.

2 Literature Review

In the domain of smart grids, foundational surveys by Ma et al. (2013) and Fang et al. (2011) discussed the technical challenges and opportunities of real-time monitoring in power systems, emphasizing the importance of data-driven decision-making. Ma et al. identified three major challenges in implementing smart grid communications: standards interoperability, cognitive access to unlicensed radio spectra, and cybersecurity. These works laid the groundwork for integrating sensor-based data with control systems, but did not focus on human-centered visualization tools or dashboard interfaces.

Previous studies in the field of industrial safety have primarily focused on traditional statistical analysis and manual reporting methods. Rathor et al. (2022) explored hybrid ensemble machine learning techniques for classifying workplace safety incidents, demonstrating the potential of predictive analytics for risk mitigation through combining Decision Tree and Random Forest algorithms with 92% accuracy. However, their work lacked interactive visualization or user-driven exploration capabilities, focusing primarily on algorithmic performance rather than user interface design.

Dashboard design principles established in previous research indicate that tabular in-

formation is superior to graphical representation for certain data types, while graphs are more effective for trends and forecasts (Few 2006). This finding has significant implications for manufacturing dashboard design, where both real-time status monitoring and trend analysis are crucial.

Tokola et al. (2016) proposed a design framework for manufacturing dashboards based on KPI surveys, establishing that different organizational levels require distinct dashboard functionalities - operational dashboards for workers, tactical dashboards for managers, and strategic dashboards for executives. Their research revealed that delivery reliability and punctuality were the most important KPIs across all organizational levels, with mobile tablet usage being the preferred platform for dashboard access. However, their study remained largely conceptual and did not evaluate interactive dashboard frameworks in real manufacturing environments.

The integration of advanced communication technologies in manufacturing has been extensively studied, with researchers emphasizing the role of wireless sensor networks and machine-to-machine communications in enabling real-time monitoring capabilities (Gungor and Lambert 2006; Fadlullah et al. 2011). The evolution toward Industry 4.0 has highlighted the need for sophisticated communication infrastructures that can handle distributed energy resources and renewable energy integration, with the Internet of Things playing a crucial role in connecting industrial systems (Zhong et al. 2017). However, most existing research focuses on the technical infrastructure rather than the visualization and user interaction aspects of these systems.

Recent developments in manufacturing analytics have shown promising results in applying machine learning techniques for safety prediction and performance monitoring. The hybrid ensemble approach demonstrated by Rathor et al. (2022) achieved significant improvements in prediction accuracy by combining multiple algorithms rather than relying on single models. This approach is particularly relevant for manufacturing environments where safety prediction accuracy is critical, complementing broader trends in data-driven smart manufacturing that emphasize deep learning applications for predictive maintenance and quality control (Wang et al. 2018). Yet the translation of these analytical capabilities into accessible dashboard interfaces remains underexplored.

In summary, prior research supports the need for real-time data analysis in both energy and safety domains but lacks comprehensive, interactive visualization platforms that enable exploratory analysis and integrated risk communication. While foundational work has established dashboard design principles and smart grid communication requirements, there remains a significant gap in developing industry-specific, interactive visualization systems that can accommodate the complex data integration needs of modern manufacturing environments. Our work bridges this gap using Plotly Dash to deliver flexible dashboards tailored for smart grid and industrial safety analytics, addressing the identified limitations in current research through practical implementation and user-centered design.

3 Methodology

3.1 Data Collection

We download the two datasets below from Kaggle:

- Smart Grid Real-Time Load Monitoring Dataset

We mainly focus on this "Smart-Grid" dataset for the project. It contains real-time load monitoring data for smart grid systems, structured as a time-series dataset with 50,000+ records recorded at 15-minute intervals. It includes key electrical parameters, renewable energy sources, environmental factors, and anomaly indicators, making it suitable for analyzing energy consumption patterns and visualization.

Table 1: Description of Attributes in the Smart-Grid Dataset

Attribute	Description
Timestamp	Time of data recording.
Voltage (V), Ampere (A), Power (kW)	Electrical measurements.
Reactive Power (kVAR), Power Factor	Energy efficiency indicators.
Solar & Wind Power (kW)	Renewable energy input.
Grid Supply (kW)	Power drawn from the grid.
Fault Indicators	Overload conditions, voltage fluctuations, transformer faults.
Environmental Factors	Temperature, humidity, and electricity price.
Predicted Load (kW)	Target variable for energy forecasting.

Its temporal and multivariate structure aligns well with our project goal of building a manufacturing KPI dashboard, as well as smart grid analytics, supporting research in real-time energy optimization and predictive modeling.

- Industrial Safety and Health Analytics Database

We utilize this "Industrial-Safety" dataset as a secondary dataset. It contains detailed records of workplace incidents and safety compliance from various industrial environments. The database is basically records of accidents from 12 different plants in 3 different countries which every line in the data is an occurrence of an accident.

Table 2: Description of Attributes in the Industrial-Safety Dataset

Attribute	Description
Data	Timestamp or time/date information.
Countries	Country where the accident occurred (anonymized).
Local	City where the manufacturing plant is located (anonymized).
Industry Sector	Sector the plant belongs to.
Accident Level	From I to VI, indicates how severe the accident was.
Potential Accident Level	Indicates how severe the accident could have been.
Genre	Whether the person is male or female.
Employee or Third Party	Whether the injured person was an employee or a third party.
Critical Risk	Description of the risk involved in the accident.
Description	Detailed account of how the accident happened.

It provides structured tabular data suitable for risk-based KPI visualization, making it valuable for analyzing safety trends and operational risk factors. Integrating this dataset will allow us to extend our dashboard beyond performance metrics to include safety monitoring, helping users track and visualize incident patterns and promote safer manufacturing operations.

3.2 Data Preprocessing

Both datasets required cleaning and transformation prior to analysis.

For the smart grid dataset:

- Columns were cleaned for unit consistency (e.g., converting power units to kilowatts).
- Irregular timestamps or gaps were filled using forward-fill or interpolation methods.
- Outliers in voltage, current, or environmental sensors were capped using IQR-based filtering.
- Aggregated features such as average power per day and renewable energy ratio were derived for visualization.

For the industrial safety data:

- Non-date records in the ‘Data’ column were filtered using regular expressions.
- Timestamp fields were converted using `pandas.to_datetime` with coercion and null-value removal.
- Categorical fields such as ‘Accident Level’, ‘Genre’, and ‘Critical Risk’ were standardized and encoded.
- Missing fields in descriptions were dropped to ensure accurate word cloud generation.

3.3 Analysis Techniques

Our project workflow involves four key stages: data processing, statistical modeling, visualization, and deployment. The tools are selected to support efficient development at each stage:

- **Python** – Used throughout for data ingestion, transformation, modeling, and simulation.
- **Pandas, NumPy** – For tabular data wrangling and numerical operations.
- **Apache Spark** – Employed for distributed processing of large-scale datasets to ensure scalability and efficiency in preprocessing and transformation.
- **Scikit-learn, statsmodels** – Provide time-series modeling and baseline regression analysis for load forecasting and risk modeling. Linear Regression and Decision Tree are used in this project.
- **Plotly Dash** – Enables interactive, web-based dashboards without requiring HTML/JavaScript.
- **Jupyter / Colab** – For iterative analysis and experimentation.
- **GitHub** – Supports collaboration and version control across iterations.

4 Results

4.1 Dashboards Overview

We design and implement two dashboards for "Smart-Grid" (Figure 4) and "Industrial-Safety" (Figure 5), both with figures and charts derived from their datasets' attributes. The original records are also displayed at the bottom.





Both dashboards allow users to interactively explore data by filtering and visualizing it through dropdown selections (Figure 6) and real-time chart updates:

- 9

- **Binary Event Distribution:** A pie chart illustrates the distribution of binary events such as *overload conditions* and *transformer faults*.
- **Environmental Correlation:** A scatter plot visualizes the correlation between *temperature*, *humidity*, and *electricity price*.
- **Industrial Safety Interface:** In the safety dashboard, users can select filters for *Accident Level*, *Country*, and *Industry Sector*, which update multiple bar plots, pie charts, and time trends. Besides, a word cloud image visualizes common (high-frequency) accident descriptions.
- **Data Table:** The filtered dataset records are presented below the graphs in a scrollable or paginated table for raw data inspection.

This unified dashboard architecture provides stakeholders with a flexible, interactive tool to simultaneously analyze operational metrics and safety records, supporting faster, data-driven decisions in energy and manufacturing domains.

The figure illustrates a three-step process for selecting a date using dropdown menus. Each step shows the state of the Year, Month, and Day selectors.

Step	Year	Month	Day
1	2016 (selected), 2017	Select Month	Select Day
2	2016	01 (selected), 02, 03	Select Day
3	2016	04	01 (selected), 02, 03

Figure 6: Date Select Dropbox

4.3 Accident Forecasting with Machine Learning

To forecast the number of industrial accidents over time, we adopt a machine learning approach leveraging historical data. Specifically, we employ regression models to predict monthly accident counts, enabling trend analysis and supporting proactive safety measures.

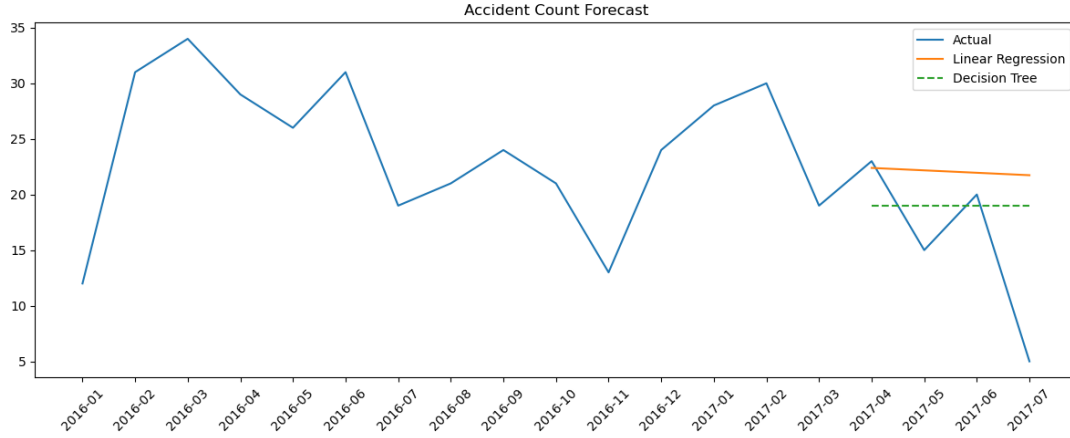


Figure 7: Accident Forecasting Results

Table 3: Evaluation Metrics of Regression Models

Model	Mean Squared Error (MSE)	R ² Score
Linear Regression	83.94	-0.798
Decision Tree Regressor	57.25	-0.226

Both models offer a simple method to forecast monthly accident trends. While the Decision Tree can capture non-linear trends better, Linear Regression provides a clear baseline. Further improvements can be made using more features and real-time data.

5 Discussion

The results show that interactive dashboards can effectively present multi-dimensional insights from both "Smart-Grid" and "Industrial-Safety" datasets. The ability to drill down by date, explore categorical patterns, and visualize textual causes enhances situational awareness.

The word cloud analysis revealed consistent patterns in safety descriptions that can inform training programs. The forecasting models demonstrated that even basic historical trends can be leveraged for proactive planning.

Compared to static dashboards in existing literature, our approach provides user-centric interactivity, real-time filtering, and full-stack deployment using open-source tools. One limitation is that the safety dataset lacks sensor data, which reduces predictive power. Moreover, the smart grid dashboard remains static rather than being fed by a live data stream.

Future improvements could include real-time ingestion, richer model integration, and UI enhancements for mobile responsiveness.

6 Conclusion

This project demonstrates the viability and utility of using Plotly Dash to create modular, interactive dashboards for industrial and energy analytics. We integrated two heterogeneous datasets, performed preprocessing, and deployed visual tools that aid users in identifying trends, causes, and critical risks.

The project reinforces that visual analytics plays a key role in operational intelligence. The dashboards not only provide static insights but also offer extensible platforms for future integration with real-time data and machine learning models.

Limitations such as lack of live data and limited feature engineering in the modeling phase were noted. Future work will focus on improving data granularity, integrating sensor streams, and enhancing the forecasting logic using hybrid models.

7 References

<https://www.finereport.com/en/data-visualization/data-visualization-dashboard.html>

<https://envelio.com/smart-grid>

<https://btsconsultant.com/courseDetails/1546>

Ma, Ruofei, et al. "Smart grid communication: Its challenges and opportunities." IEEE transactions on Smart Grid 4.1 (2013): 36-46.

Fang, Xi, et al. "Smart grid—The new and improved power grid: A survey." IEEE communications surveys & tutorials 14.4 (2011): 944-980.

Rathor, Ketan, et al. "A Detailed View on Industrial Safety and Health Analytics using Machine Learning Hybrid Ensemble Techniques." 2022 International Conference on Edge Computing and Applications (ICECAA). IEEE, 2022.

Few, Stephen. "Information dashboard design." (2006).

Tokola, Henri, et al. "Designing manufacturing dashboards on the basis of a Key Performance Indicator survey." Procedia Cirp 57 (2016): 619-624.

Gungor, Vehbi C., and Frank C. Lambert. "A survey on communication networks for electric system automation." Computer Networks 50.7 (2006): 877-897.

Fadlullah, Zubair Md, et al. "Toward intelligent machine-to-machine communications in smart grid." IEEE Communications Magazine 49.4 (2011): 60-65.

Zhong, Ray Y., et al. "Intelligent manufacturing in the context of industry 4.0: a review." Engineering 3.5 (2017): 616-630.

Wang, Jinjiang, et al. "Deep learning for smart manufacturing: Methods and applications." Journal of manufacturing systems 48 (2018): 144-156.

<https://www.kaggle.com/datasets/ziya07/smart-grid-real-time-load-monitoring-dataset/data>

<https://www.kaggle.com/datasets/ihmstefanini/industrial-safety-and-health-analytics-database>